

```
1 #include <stdio.h>
2
3 struct Review {
4     char reviewer[50];
5     int rating;
6 };
7
8 struct Book {
9     char title[100];
10    char author[50];
11    struct Review reviews[5];
12    int reviewCount;
13 };
14
15 void inputBook(struct Book *b) {
16     printf("Enter book title: ");
17     scanf("%[^\n]%*c", b->title);
18     printf("Enter author name: ");
19     scanf("%[^\n]%*c", b->author);
20
21     printf("Enter number of reviews: ");
```

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Book {
5     char title[100];
6     char author[50];
7     int year;
8 };
9
10 void inputBooks(struct Book b[], int n) {
11     for (int i = 0; i < n; i++) {
12         printf("\nBook %d:\n", i + 1);
13         printf("Enter title: ");
14         scanf("%[^\n]*c", b[i].title);
15         printf("Enter author: ");
16         scanf("%[^\n]*c", b[i].author);
17         printf("Enter publication year: ");
18         scanf("%d", &b[i].year);
19         getchar();
20     }
21 }
```

```
1 #include <stdio.h>
2
3 struct Student {
4     char name[50];
5     int id;
6     float grades[10];
7     int numGrades;
8 };
9
10 void calculateAverage(struct Student s) {
11     float sum = 0, average;
12     int i;
13
14     for (i = 0; i < s.numGrades; i++) {
15         sum += s.grades[i];
16     }
17
18     average = sum / s.numGrades;
19
20     printf("\nStudent Name: %s", s.name);
21     printf("\nStudent ID: %d", s.id);
```

```
1 #include <stdio.h>
2
3 struct Car {
4     char make[30];
5     char model[30];
6     int year;
7     float price[3];    // 0 = new, 1 = used, 2 = damaged
8 };
9
10 void findPrice(struct Car cars[], int n) {
11     char make[30], condition[20];
12     int i, found = 0;
13
14     printf("Enter car make: ");
15     scanf("%99s", make);
16
17     printf("Enter condition (new/used/damaged): ");
18     scanf("%99s", condition);
19
20     for (i = 0; i < n; i++) {
21         int j = -1;
```

```
1 #include <stdio.h>
2
3 struct Earnings {
4     float Q1;
5     float Q2;
6     float Q3;
7     float Q4;
8     float department[5];
9 };
10
11 void calculateTotal(struct Earnings e) {
12     float total = 0;
13     int i;
14
15     total = e.Q1 + e.Q2 + e.Q3 + e.Q4;
16
17     for (i = 0; i < 5; i++) {
18         total += e.department[i];
19     }
20
21     printf("\nTotal Earnings = %.2f\n", total);
```

```
1 #include <stdio.h>
2
3 struct Movie {
4     char title[50];
5     char director[50];
6     char actor[50];
7     float ratings[5];
8 };
9
10 void displayMovie(struct Movie movies[], int n) {
11     char search[50];
12     int i, j, found = 0;
13     float sum, average;
14
15     printf("Enter movie title: ");
16     scanf("%[^\\n]", search);
17
18     for (i = 0; i < n; i++) {
19         j = 0;
20
21         while (search[j] == movies[i].title[j] && search[j] != '\\0') {
```

```
1 #include <stdio.h>
2
3 struct Weather {
4     float temperature;
5     float humidity;
6     float windSpeed;
7     float precipitation[7];
8 };
9
10 void calculateAverage(struct Weather w) {
11     float sum = 0, average;
12     int i;
13
14     for (i = 0; i < 7; i++) {
15         sum += w.precipitation[i];
16     }
17
18     average = sum / 7;
19
20     printf("\nAverage Precipitation: %.2f mm\n", average);
21 }
```



```
1 #include <stdio.h>
2
3 struct Product {
4     char name[50];
5     float price;
6     int quantity[12];
7 };
8
9 void calculateRevenue(struct Product p[], int n) {
10     int i, j;
11     int totalQuantity;
12     float revenue;
13
14     for (i = 0; i < n; i++) {
15         totalQuantity = 0;
16
17         for (j = 0; j < 12; j++) {
18             totalQuantity += p[i].quantity[j];
19         }
20
21         revenue = p[i].price * totalQuantity;
```



```
1 #include <stdio.h>
2
3 struct Transaction {
4     char type[20];
5     float amount;
6     char date[20];
7     char descriptions[5][100];
8 };
9
10 void displayDescriptions(struct Transaction t) {
11     int i;
12
13     printf("\nTransaction Type: %s", t.type);
14     printf("\nAmount: %.2f", t.amount);
15     printf("\nDate: %s", t.date);
16
17     printf("\nDescriptions:\n");
18     for (i = 0; i < 5; i++) {
19         printf("%d. %s\n", i + 1, t.descriptions[i]);
20     }
21 }
```

```
1 #include <stdio.h>
2
3 struct Employee {
4     char name[50];
5     float salary;
6     float bonus[4];
7 };
8
9 void calculateEarnings(struct Employee e[], int n) {
10     int i, j;
11     float totalBonus, totalEarnings;
12
13     for (i = 0; i < n; i++) {
14         totalBonus = 0;
15
16         for (j = 0; j < 4; j++) {
17             totalBonus += e[i].bonus[j];
18         }
19
20         totalEarnings = e[i].salary + totalBonus;
21     }
```

```
1 #include <stdio.h>
2
3 struct Person {
4     char name[50];
5     int age;
6     char address[100];
7 };
8
9 void displayPerson(struct Person *p) {
10     printf("\nName: %s", p->name);
11     printf("\nAge: %d\n", p->age);
12 }
13
14 int main() {
15     struct Person p;
16
17     printf("Enter name: ");
18     scanf(" %[^\\n]", p.name);
19
20     printf("Enter age: ");
21     scanf("%d", &p.age);
```

```
1 #include <stdio.h>
2
3 struct Person {
4     char name[50];
5     int age;
6     char address[100];
7 };
8
9 void modifyPerson(struct Person *p) {
10     printf("\nEnter new name: ");
11     scanf(" %[^\\n]", p->name);
12
13     printf("Enter new age: ");
14     scanf("%d", &p->age);
15
16     printf("Enter new address: ");
17     scanf(" %[^\\n]", p->address);
18 }
19
20 int main() {
21     struct Person p;
```

```
1 #include <stdio.h>
2
3 struct Student {
4     char name[50];
5     int id;
6     float grades[5];
7 };
8
9 void calculateAverage(struct Student *s) {
10     float sum = 0, average;
11     int i;
12
13     for (i = 0; i < 5; i++) {
14         sum += s->grades[i];
15     }
16
17     average = sum / 5;
18
19     printf("\nStudent Name: %s", s->name);
20     printf("\nStudent ID: %d", s->id);
21     printf("\nAverage Grade: %.2f\n", average);
```

```
1 #include <stdio.h>
2
3 struct Array {
4     int numbers[10];
5     int size;
6 };
7
8 void findMaxMin(struct Array *a) {
9     int max, min, i;
10
11     max = a->numbers[0];
12     min = a->numbers[0];
13
14     for (i = 1; i < a->size; i++) {
15         if (a->numbers[i] > max)
16             max = a->numbers[i];
17
18         if (a->numbers[i] < min)
19             min = a->numbers[i];
20     }
21 }
```

```
1 #include <stdio.h>
2
3 struct Transaction {
4     char type[20];
5     float amount;
6     char date[20];
7     char descriptions[5][100];
8 };
9
10 void displayTransaction(struct Transaction *t) {
11     int i;
12
13     printf("\nTransaction Type: %s", t->type);
14     printf("\nAmount: %.2f", t->amount);
15     printf("\nDate: %s", t->date);
16
17     printf("\nDescriptions:\n");
18     for (i = 0; i < 5; i++) {
19         printf("%d. %s\n", i + 1, t->descriptions[i]);
20     }
21 }
```



```
1 #include <stdio.h>
2
3 struct Product {
4     char name[50];
5     float price;
6     int quantity;
7 };
8
9 void calculateRevenue(struct Product *p, int n) {
10     int i;
11     float revenue;
12
13     for (i = 0; i < n; i++) {
14         revenue = p[i].price * p[i].quantity;
15
16         printf("\nProduct: %s", p[i].name);
17         printf("\nPrice: %.2f", p[i].price);
18         printf("\nQuantity Sold: %d", p[i].quantity);
19         printf("\nTotal Revenue: %.2f\n", revenue);
20     }
21 }
```

```
1 #include <stdio.h>
2
3 struct Earnings {
4     float Q1;
5     float Q2;
6     float Q3;
7     float Q4;
8     float department[5];
9 };
10
11 void calculateTotal(struct Earnings *e) {
12     float total;
13     int i;
14
15     total = e->Q1 + e->Q2 + e->Q3 + e->Q4;
16
17     for (i = 0; i < 5; i++) {
18         total += e->department[i];
19     }
20
21     printf("\nTotal Earnings = %.2f\n", total);
```

```
1 #include <stdio.h>
2
3 struct Weather {
4     float temperature;
5     float humidity;
6     float windSpeed;
7     float precipitation;
8 };
9
10 void updateWeather(struct Weather *w) {
11     printf("Enter temperature (°C): ");
12     scanf("%f", &w->temperature);
13     printf("Enter humidity (%%): ");
14     scanf("%f", &w->humidity);
15     printf("Enter wind speed (km/h): ");
16     scanf("%f", &w->windSpeed);
17     printf("Enter precipitation (mm): ");
18     scanf("%f", &w->precipitation);
19 }
20
```

```
1 #include <stdio.h>
2
3 struct Review {
4     char reviewer[50];
5     int rating;
6 };
7
8 struct Book {
9     char title[100];
10    char author[50];
11    struct Review reviews[5];
12    int reviewCount;
13 };
14
15 void displayBook(struct Book *b) {
16     printf("\n--- Book Details ---\n");
17     printf("Title: %s\nAuthor: %s\n", b->title, b->author);
18     printf("\nReviews:\n");
19     for (int i = 0; i < b->reviewCount; i++) {
20         printf("%s rated %d/5\n", b->reviews[i].reviewer, b->reviews[i].rating);
21     }
```

```
1 #include <stdio.h>
2
3 struct Car {
4     char make[50];
5     char model[50];
6     int year;
7     float price[3]; // new, used, damaged
8 };
9
10 void inputCar(struct Car *c) {
11     printf("Enter make: ");
12     scanf("%s", c->make);
13     printf("Enter model: ");
14     scanf("%s", c->model);
15     printf("Enter year: ");
16     scanf("%d", &c->year);
17     printf("Enter price (new, used, damaged): ");
18     for (int i = 0; i < 3; i++)
19         scanf("%f", &c->price[i]);
20 }
```